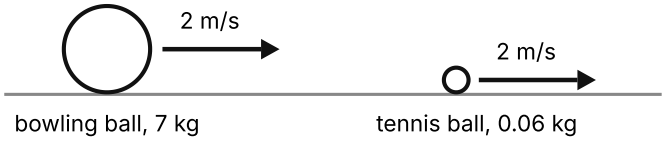
Mini-Lesson 1: Momentum (p = mv)

AP Physics 1 · Unit 4 · Do this with the mini-lesson before class Thu 11/12. Fill in each box when the lesson shows its orange tag.

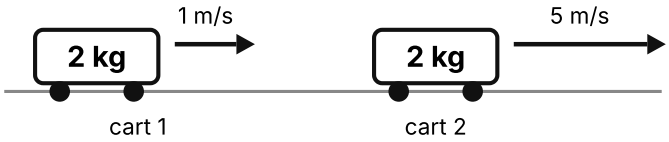
Scan with your iPad camera to open the lesson

Name _____ Period _____

A What is momentum?



Harder to stop: _____ because it has more _____.



Harder to stop: _____ because it has more _____.

Momentum measures how hard it is to _____ a moving object. It depends on _____ and _____.

B The equation

p = _____ × _____

In words: momentum = _____ × _____

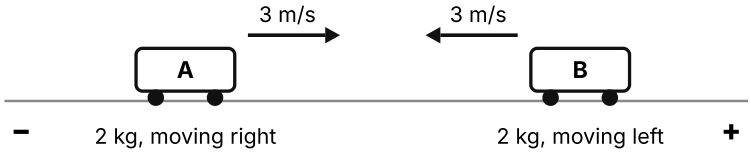
Units: _____ (the symbol for momentum is **p**)

If mass doubles, p _____.

If velocity doubles, p _____.

Slider: with a negative velocity, the momentum arrow points _____.

C Direction matters (right is +)



p_A = _____ p_B = _____

Momentum is a _____: it points the same direction as _____.

The **sign** tells the _____. The **number** tells the _____.

Car driving west (east is +): p is _____.

D Worked example (east is +)

Object	m	v	p = mv
Halfback (east)	80 kg	+9 m/s	_____
Car	900 kg	−20 m/s	_____
Freshman, walking	40 kg	+2 m/s	_____

Hardest to stop: _____. Easiest: _____.

Compare the _____, not the sign.

E Check questions: show your work

1. 0.15 kg baseball moving left at 30 m/s (right is +)

work:

p = _____

2. 3000 kg truck at 10 m/s vs. 1000 kg car at 25 m/s

work:

More p: _____

3. v doubles, m is cut in half. p = (½m)(2v) → p _____

Big idea in one sentence:

_____

_____